

1) AVL Tree

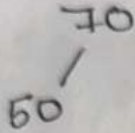
keys: 70, 50, 30, 90, 80, 130, 120 & deletion of 120 & 30.

steps:

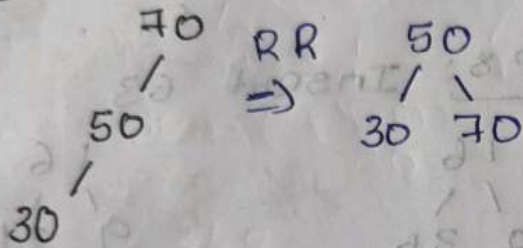
Insert 80

step 1: 70

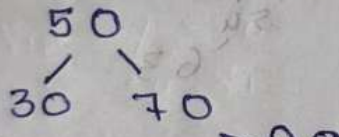
step 2: Insert 50



step 3: Insert 30

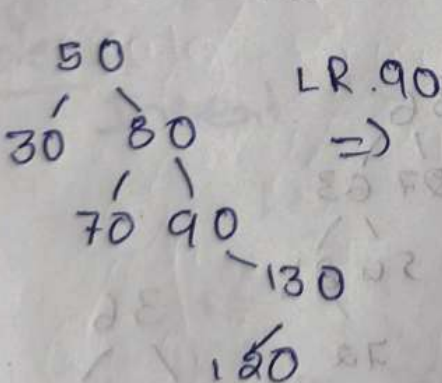


step 4: Insert 90

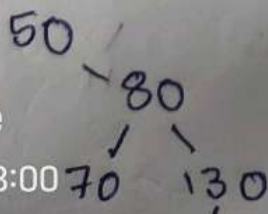


step 5

Insert 120

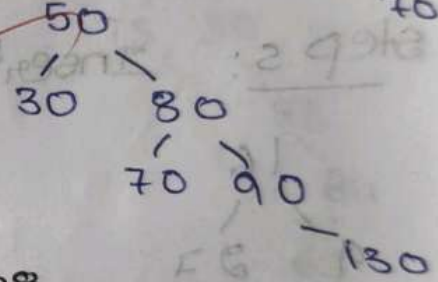


step 9: Delete 30



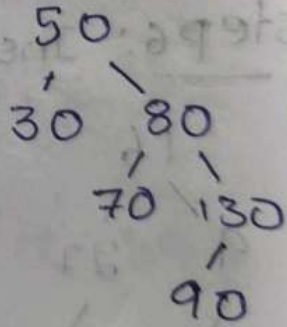
step 6

Insert 130



step 8

Delete 120



AVL tree

keys: 16, 27, 9, 11, 36, 54, 81, 63, 72.

step 1:

16

step 2:

16
27

step 3:

16
9 27

step 4: Insert 11

16
9 27
11

step 5: Insert 36

16
9 27
11 36

step 6: Insert 54

16
9 27
11 36
54

At 27 → RR

Rotation at 27

step 7: Insert 81

16
9 36
11 27 54
81

Left rotation at 54

16
9 36
11 27 81
54

step 8: Insert 63

16
9 36
11 27 81
54 63

step 9: Insert 72

16
9 36
11 27 63
54 81
72

3) Binary search tree

key s: 11, 25, 10, 15, 39, 58, 84, 65, 78

step 1: Insert 11

step 2:

Insert 25

step 3:

Insert 10

step 4:

Insert 15

step 5:

Insert 39

step 6:

Insert 58

step 7:

Insert 84

step 8:

Insert 65

step 9:

Insert 78

12/08/20

4) Binary search tree

Key: 9, 27, 50, 15, 2, 21, 36

step 1:

9

step 2:

9
└ 27

step 3:

9
└ 27
 └ 50

step 4:

9
└ 27
 └ 15 50

step 5:

9
└ 27
 └ 15 50
 └ 21

step 6:

9
└ 27
 └ 15 50
 └ 21

step 7:

9
└ 27
 └ 15 50
 └ 21 36

Final BST :

9
└ 27
 └ 15 50
 └ 21 36